connection. The solution is to encode the HTTP headers along with the request data into an encrypted *request object*, as shown in listing 13.14.

> **DEFINITION** A ***request object*** is an API request that is encapsulated as a single data object that can be encrypted and authenticated as one element. The request object captures the data in the request as well as headers and other metadata required by the request.

In this example the headers are encoded as a CBOR map, which is then combined with the request body and an indication of the expected HTTP method to create the overall request object. The entire object is then encrypted and authenticated using NaCl's `CryptoBox` functionality. OSCORE, discussed in section 13.1.4, is an example of an end-to-end protocol using request objects. The request objects in OSCORE are CoAP messages encrypted with COSE.

> **TIP** Full source code for this example is provided in the GitHub repository accompanying the book at https://github.com/NeilMadden/apisecurityinaction/blob/chapter13-end/natter-api/src/main/java/com/manning/apisecurityinaction/ReplayProtectionExample.java.

Listing 13.14 Encoding HTTP headers into a request object

```
var revisionEtag = "42";    #A
```